

Reproducing and Extending a Dynamical Model of Dopamine Regulation During Levodopa Therapy

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AMATH 382 — Final Project

April 15, 2025

Abstract

Parkinson’s disease (PD) results from the gradual loss of dopaminergic neurons in the substantia nigra pars compacta (SNc), and while levodopa (LD) therapy remains the gold-standard treatment, its efficacy deteriorates over time as the therapeutic window narrows and dyskinesias emerge. This project aims to reproduce key computational results from the ODE-based models of Best et al. [1] and Reed et al. [2], which together describe dopamine synthesis, storage, release, reuptake, autoreceptor regulation, and the effects of levodopa dosing under progressive neurodegeneration. The dopaminergic terminal system was re-implemented in Python, and results were obtained for steady-state homeostasis, the passive stabilization of extracellular dopamine, time courses during LD dosing at several levels of cell loss, and quantification of the therapeutic window. As a novel extension, extracellular dopamine was coupled to a simplified model of neuronal excitability to show how degeneration-driven DA fluctuations propagate into altered gain dynamics in downstream striatal circuits, providing a conceptual link between biochemical regulation and circuit-level dysfunction.

1 Introduction and Motivation

Parkinson’s disease is among the most prevalent neurodegenerative disorders worldwide, affecting over 6 million people globally [3]. Its hallmark motor symptoms (such as tremor, bradykinesia, and rigidity) arise from progressive degeneration of dopaminergic neurons in the SNc, which project to the striatum and modulate motor control through the basal ganglia [4]. The dopamine precursor levodopa, which unlike dopamine itself can cross the blood-brain barrier and is then converted to DA in the brain, has been the primary pharmacological treatment since the late 1960s [5].

Despite early success, long-term LD therapy brings a well-documented set of complications. The therapeutic time window, also known as the duration of adequate symptom relief following a single dose, gradually shortens as the disease progresses and eventually approaches the plasma half-life of levodopa at roughly 90 minutes [6, 7]. Patients begin to experience wearing-off episodes, on-off motor fluctuations, and levodopa-induced dyskinesias (LID)—involuntary, erratic move-

ments that arise from overstimulation of dopamine receptors during peak LD levels—all of which substantially reduce quality of life and limit the practicality of continued treatment [8].

Reed et al. [2] developed a mathematical model that offers a mechanistic explanation for these complications. Building on the dopaminergic terminal model of Best et al. [1], they demonstrated that the interplay between DA release, DAT-mediated reuptake, and the progressive loss of SNc terminals accounts for both the passive stabilization of extracellular dopamine in early disease and the collapse of the therapeutic window in advanced stages.

1.1 Purpose of the Model

By the time Parkinson’s disease becomes symptomatic, the striatum has typically lost upwards of 60–80% of its dopaminergic input [9]. The system compensates so well for so long that the damage is essentially invisible until it isn’t. Understanding *why* that compensation eventually fails, and why levodopa stops working reliably as the disease progresses, requires more than a general sense of the biology. A mathematical model is useful here precisely because it lets you do that: vary one parameter at a time, track all nine state variables simultaneously, and watch how the system’s behaviour changes across different levels of degeneration in a way that would be impractical or ethically challenging in animal models.

1.2 Project Scope

This project focuses on reproducing the dopaminergic terminal subsystem from Best et al. [1] and the levodopa-related analyses from Reed et al. [2]. The specific results reproduced are:

1. Steady-state homeostasis of the DA terminal,
2. Substrate inhibition of tyrosine hydroxylase (TH) by tyrosine,
3. Passive stabilization of extracellular DA under progressive cell loss,
4. Extracellular DA time courses during LD dosing at varying levels of SNc degeneration, and
5. The therapeutic time window as a function of disease progression.

As a novel extension, the extracellular DA dynamics are coupled to a simplified neuronal gain model to investigate how dopamine fluctuations may alter downstream excitability in striatal circuits.

It is worth noting that, in this model, we are concerned with how cells regulate dopamine at the terminal level; levodopa enters the system as an exogenous input that supplies additional substrate for conversion to dopamine, and the model tracks how the dopaminergic terminals process and respond to that input rather than modelling the pharmacokinetics of the pill itself.

2 The Published Model

2.1 Overview of the Dopaminergic Terminal

The model developed by Best et al. [1] represents a single dopaminergic neuron terminal in the striatum as a system of nine coupled ODEs. The variables track concentrations of key bio-

chemical species: dihydrobiopterin (bh_2), tetrahydrobiopterin (bh_4), intracellular tyrosine (tyr), L-DOPA (ldopa), cytosolic dopamine (cda), vesicular dopamine (vda), extracellular dopamine (eda), homovanillic acid (hva), and a tyrosine pool (tyrpool) representing competing cellular uses of tyrosine.

To put briefly, tyrosine is transported into the terminal from blood, converted to L-DOPA by tyrosine hydroxylase (TH) using bh_4 as a cofactor (the rate-limiting step), then further converted to cytosolic dopamine by AADC. The vesicular monoamine transporter (VMAT) packages cytosolic DA into vesicles, from which it is released into the extracellular space upon neuronal firing. DATs rapidly clear extracellular DA back into the cytosol, and the remainder is lost through enzymatic catabolism or diffusion. The core equations for the three most clinically relevant compartments are:

$$\frac{d[\text{cda}]}{dt} = V_{\text{AADC}}([\text{ldopa}]) - V_{\text{MAT}}([\text{cda}], [\text{vda}]) + V_{\text{DAT}}([\text{eda}]) - k_{\text{cda}}^{\text{cat}} \cdot [\text{cda}] \quad (1)$$

$$\frac{d[\text{vda}]}{dt} = V_{\text{MAT}}([\text{cda}], [\text{vda}]) - \text{fire}(t) \cdot [\text{vda}] \quad (2)$$

$$\frac{d[\text{eda}]}{dt} = \text{fire}(t) \cdot [\text{vda}] - V_{\text{DAT}}([\text{eda}]) - V_{\text{eda}}^{\text{cat}}([\text{eda}]) - k_{\text{rem}} \cdot [\text{eda}] \quad (3)$$

where V_{AADC} , V_{MAT} , and V_{DAT} are Michaelis–Menten velocity functions, $\text{fire}(t)$ is the firing-dependent release rate, and k_{rem} captures clearance by diffusion and glial uptake.

2.2 Key Regulatory Mechanisms

Three homeostatic mechanisms are central to the system’s robustness, and it is worth understanding each one before looking at what happens when degeneration starts to erode them.

1. **Substrate inhibition of TH by tyrosine.** At high tyrosine concentrations, tyrosine binds non-enzymatically to TH and actually slows it down [10], which sounds counterintuitive but is what creates the flat plateau in synthesis rate around the normal operating point. The full TH velocity is:

$$V_{\text{TH}} = \underbrace{\frac{0.56}{1 + ([\text{tyr}]/K_i^{\text{tyr}})^2}}_{\text{substrate inhibition}} \cdot \underbrace{f_{\text{auto}}([\text{eda}])}_{\text{autoreceptors}} \cdot \underbrace{\frac{V_{\text{max}} \cdot [\text{tyr}] \cdot [\text{bh}_4]}{([\text{tyr}] + K_{\text{tyr}})([\text{bh}_4] + K_{\text{bh}_4})(1 + [\text{cda}]/K_i^{\text{cda}})}}_{\text{Michaelis–Menten with end-product inhibition}} \quad (4)$$

The practical consequence is that the large meal-driven swings in blood tyrosine that can roughly double brain tyrosine levels have almost no effect on how much dopamine gets made.

2. **D2 autoreceptor feedback.** Extracellular dopamine feeds back onto the terminal’s own D2 autoreceptors, suppressing TH activity when $[\text{eda}]$ climbs and releasing that suppression when it drops. This negative feedback loop is what keeps extracellular DA relatively stable even when firing rate or enzyme expression changes, and it is one of the main reasons the system can compensate so effectively in early-stage degeneration.
3. **DAT-mediated reuptake.** The dopamine transporters recapture extracellular DA back

into the cytosol with high affinity ($K_m = 0.2 \mu\text{M}$), keeping extracellular concentrations in the low nanomolar range under normal conditions. Critically, DAT capacity scales with the number of surviving terminals, so as f decreases, reuptake falls in parallel with release. This is exactly what drives passive stabilization.

2.3 Extension to Levodopa Therapy (Reed et al., 2012)

Reed et al. [2] extended the Best et al. model in two important ways. First, they added a mechanism for levodopa entry: LD is transported across the blood-brain barrier via the L-amino acid transporter, where it competes with tyrosine and tryptophan for uptake, and is then converted to dopamine by AADC in both dopaminergic and serotonergic neurons. The serotonergic component matters particularly in advanced PD, where serotonergic neurons become the dominant source of striatal DA release during LD dosing. Unlike dopaminergic terminals, they lack autoreceptor-mediated feedback, so the DA they produce from exogenous LD is released in an essentially unregulated fashion—giving rise to the characteristic large, poorly-buffered DA surge that then decays as serum LD clears. Second, they introduced a degeneration parameter $f \in [0, 1]$ representing the fraction of SNc cells remaining, which scales both release capacity and DAT-mediated reuptake. The removal rate is modelled as $k_{\text{rem}} = k_0/f$, capturing the increasing probability that released DA simply diffuses out of the striatum as terminal density falls. As f decreases, baseline extracellular DA remains relatively stable through *passive stabilization* [11, 12], but the response to LD doses becomes progressively more dysregulated as the system loses the buffering capacity it once had.

2.4 Strengths and Limitations

The model’s main strengths are its grounding in measured biochemical parameters (all K_m and V_{max} values are drawn from the experimental literature), its mechanistic transparency, and its ability to generate clinically testable predictions about passive stabilization, therapeutic window shortening, and the role of serotonergic neurons in dyskinesias. Limitations include the deterministic formulation (no stochastic fluctuations), a well-mixed extracellular space approximation that ignores spatial diffusion gradients, the absence of downstream basal ganglia electrophysiology, and no modelling of synaptic plasticity or long-term adaptive changes that contribute to dyskinesia development.

3 Model Re-Implementation

3.1 Implementation Details

The dopaminergic terminal ODE system was re-implemented in Python using `scipy.integrate.solve_ivp` with the BDF (Backward Differentiation Formula) method, which is appropriate for the stiff kinetics arising from the wide separation in timescales between fast extracellular clearance and slow bipterin cycling. All parameter values were taken directly from Table 2 of Best et al. [1] and Tables A2–A5 of Reed et al. [2]; selected key parameters are listed in Table 1.

Table 1: Selected model parameters and their sources.

Parameter	Description	Value	Source
$K_m^{\text{TH,tyr}}$	TH affinity for tyrosine	46 μM	[1]
$K_i^{\text{TH,tyr}}$	TH substrate inhibition	160 μM	[10]
$K_i^{\text{TH,cda}}$	TH end-product inhibition	110 μM	[1]
K_m^{AADC}	AADC affinity for L-DOPA	130 μM	[1]
K_m^{MAT}	VMAT affinity for DA	3 μM	[1]
K_m^{DAT}	DAT affinity for DA	0.2 μM	[1]
k_{rem}	Removal rate (baseline)	400 /hr	[1]
$K_m^{\text{BBB,LD}}$	BBB affinity for LD	32 μM	[2]

The nine differential equations follow the biochemical scheme described in Section 2.1, with velocities governed by Michaelis–Menten kinetics, competitive inhibition, and autoreceptor feedback; the full system is listed in Appendix A (Section). VMAT includes bidirectional transport (forward packaging and vesicular backleak with rate constant $k_{\text{out}} = 40/\text{hr}$). For LD dose simulations, serum LD was modelled with an approximately 90-minute pharmacokinetic half-life consistent with clinical data [13], and the degeneration parameter scales the removal rate as $k_{\text{rem}} = k_0/f$.

3.2 Steady-State Verification

The model was first integrated to steady state under tonic firing conditions (~ 4 Hz) without any levodopa input. Table 2 summarizes the resulting concentrations compared to the published values from Best et al. [1]. The most clinically important variable, extracellular DA, agrees well (2.6 nM computed vs. 2.0 nM published). Cytosolic and vesicular DA are in the correct physiological range, and the quantitative discrepancies in bipterin species (bh_2 , bh_4) reflect sensitivity to the DRR (dihydrobiopterin reductase) cycling parameters and the exact formulation of the autoreceptor function, which are the most numerically sensitive parts of the system.

Table 2: Steady-state concentrations from the re-implemented model vs. published values from Best et al. [1].

Variable	This model (μM)	Published (μM)	Agreement
bh_2	3.26	41.0	Qualitative
bh_4	356.7	319.0	Qualitative
tyr	114.9	126.0	Good
ldopa	0.60	0.36	Reasonable
cda	4.46	2.65	Reasonable
vda	103.3	81.0	Reasonable
eda	0.0026	0.0020	Good
hva	12.9	7.68	Reasonable
tyrpool	861.8	952.0	Good

The qualitative behaviour under perturbation is preserved across all experiments below, which is the more important criterion for the analysis that follows.

4 Reproduced Results

All code used to implement the model and generate the figures in this section is available in the project repository at https://github.com/nahalhz/AMATH-382-Final-Project/blob/main/finalProject_nhabibiz.ipynb.

4.1 Substrate Inhibition of TH by Tyrosine

Figure 1 shows TH velocity as a function of tyrosine concentration, comparing the substrate-inhibited and standard Michaelis–Menten cases. Consistent with Best et al. [1] (their Figure 2), the inhibited curve plateaus and then declines at high tyrosine, while the uninhibited version continues to rise. The plateau falls right around the normal intracellular tyrosine range of roughly 115–126 μM , so the large meal-driven fluctuations in blood tyrosine that can double brain tyrosine levels [14] have minimal impact on dopamine synthesis rate. This is one of the central homeostatic properties of the system that allows it to operate relatively stably despite variable dietary input.

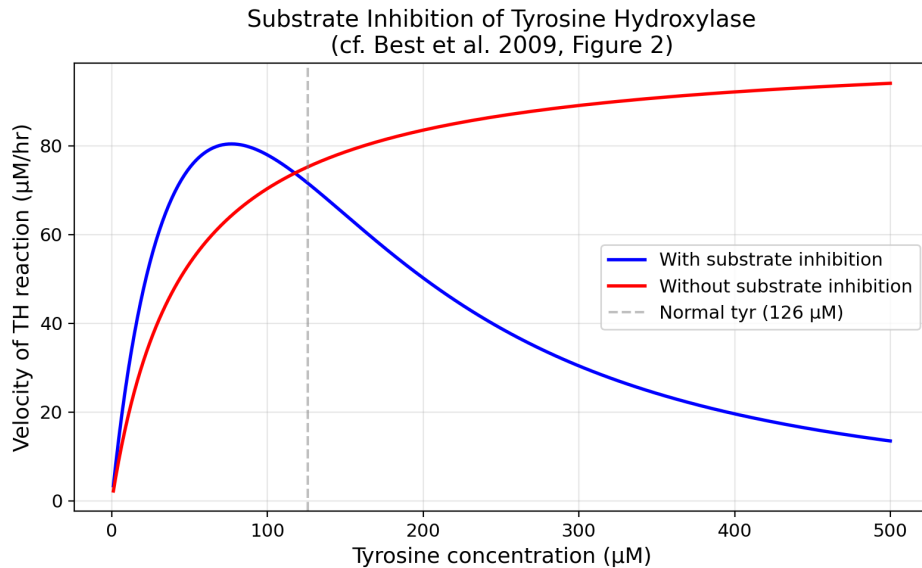


Figure 1: TH reaction velocity as a function of tyrosine concentration with and without substrate inhibition. The flat plateau near the normal operating point (115–126 μM) insulates dopamine synthesis from dietary tyrosine fluctuations (cf. Best et al. [1], Figure 2).

4.2 Autoreceptor Homeostasis

Figure 2 shows steady-state extracellular DA as a function of firing rate (expressed as a percentage of normal). The D2 autoreceptor feedback compresses the response: extracellular DA rises only modestly even when firing rate doubles. This dampening is what allows the system to buffer normal fluctuations in neuronal activity while still permitting large, sustained signals—such as reward-related burst firing—to produce biologically meaningful dopamine transients.

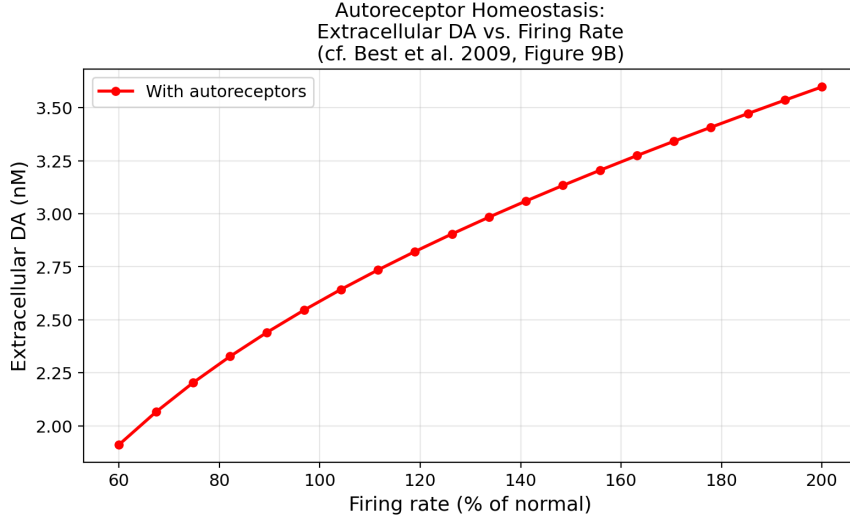


Figure 2: Steady-state extracellular DA vs. firing rate, demonstrating autoreceptor-mediated homeostasis. Even large changes in firing produce relatively modest changes in extracellular DA (cf. Best et al. [1], Figure 9B).

4.3 Passive Stabilization of Extracellular DA

Figure 3 reproduces the passive stabilization result from Reed et al. [2] (their Figure 6). As SNc cells die (f decreasing toward zero), extracellular DA remains remarkably stable until roughly 80% of cells have been lost ($f < 0.2$), after which it begins a steeper decline. This phenomenon arises because both DA release and DAT-mediated reuptake scale with f : fewer terminals mean less DA released, but also fewer transporters available to clear it, so the net extracellular concentration stays roughly constant and the system effectively hides the progressive cell loss from the outside world until it reaches a tipping point. This mechanistically explains why Parkinsonian symptoms do not appear until 60–80% of SNc neurons have degenerated [9], and it also sets the stage for why the system later becomes so fragile to perturbations like levodopa dosing.

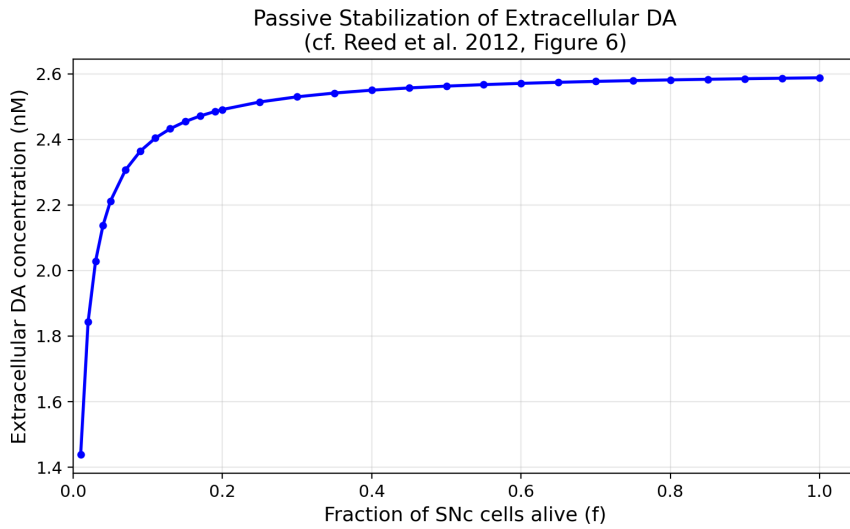


Figure 3: Passive stabilization: extracellular DA remains near baseline until more than 80% of SNc cells have been lost (cf. Reed et al. [2], Figure 6).

4.4 Extracellular DA Time Courses During LD Dosing

Figure 4 shows extracellular DA in the striatum during a levodopa dose across different values of the degeneration parameter f . For a normal individual ($f = 1$), DA rises moderately and is well-buffered, staying elevated for many hours. As degeneration progresses to $f = 0.2$ and $f = 0.1$, the DA peaks are actually higher—this happens because there are fewer DATs available to reuptake the DA that is being produced from the exogenous LD, so the DA that does get released accumulates to a greater degree in the extracellular space. For severe degeneration ($f = 0.05$ and $f = 0.01$), the peaks fall again, because the increasing removal rate $k_{\text{rem}} = k_0/f$ captures the fact that DA molecules are far more likely to simply diffuse out of the striatum when terminal density is very low rather than being recaptured.

The dashed therapeutic threshold is set halfway between the normal baseline DA level and the $f = 0.2$ baseline. The duration each curve spends above this threshold defines the therapeutic window for that disease stage.

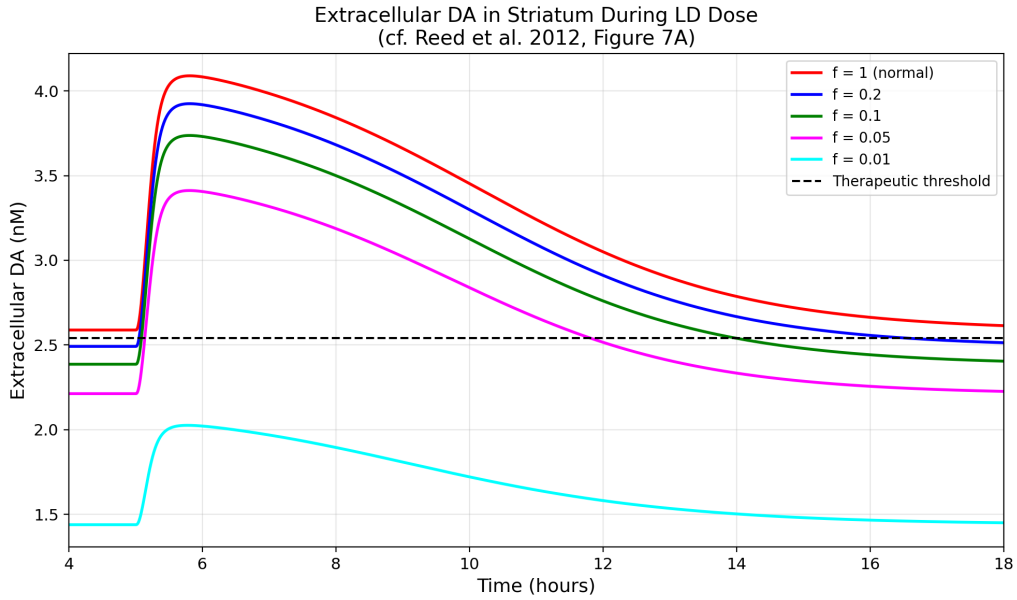


Figure 4: Extracellular DA time courses during an LD dose for five levels of SNc degeneration. The therapeutic window (time above the dashed threshold) progressively shortens with disease severity (cf. Reed et al. [2], Figure 7A).

4.5 Therapeutic Time Window

Figure 5 quantifies the therapeutic window across values of f . Starting from approximately 11.5 hours at $f = 0.2$, the window steadily shortens to near zero at $f = 0.01$. At intermediate degeneration ($f = 0.02$), the window is about 2.5 hours—close to the plasma half-life of LD itself—which is consistent with clinical observations in advanced PD [7, 8]. This shortening reflects two factors working together: the reduced capacity of the remaining terminals to store and buffer the DA arriving from serotonergic neurons, and the increasing probability that released DA will be lost from the system entirely rather than recaptured.

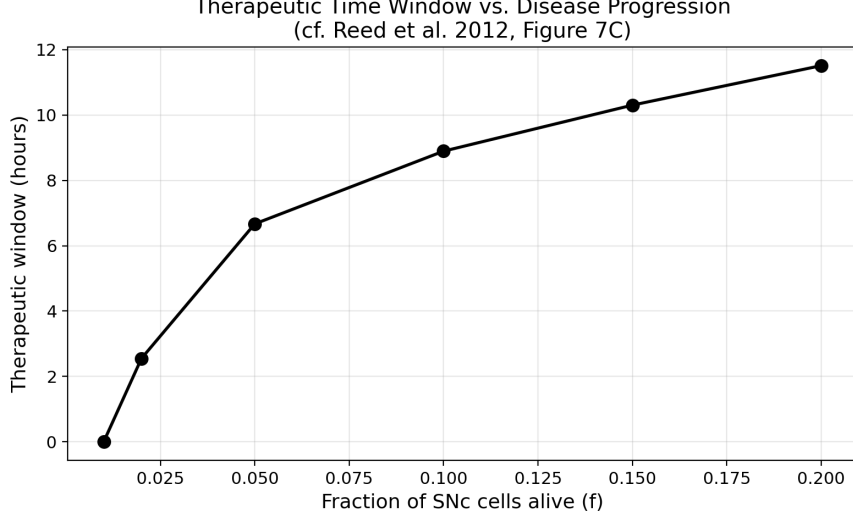


Figure 5: Therapeutic time window as a function of remaining SNc cell fraction f . The window shrinks to near the LD plasma half-life (~ 2 hours) as PD advances (cf. Reed et al. [2], Figure 7C).

5 Novel Extension: Dopamine-Modulated Neural Excitability

5.1 Motivation and Model Formulation

The published model ends at extracellular dopamine concentration and does not connect the biochemical dynamics to the neural circuit dysfunction that actually produces Parkinsonian symptoms. Parkinson’s disease manifests as a motor disorder not because dopamine molecules are missing in a vacuum, but because their absence—and the erratic surges caused by levodopa—dysregulates downstream striatal circuit activity. To build a bridge between the two levels of description, I couple the extracellular DA output to a simplified model of dopamine-modulated neural excitability.

The downstream striatal neuron is described by a leaky integrator whose response gain is a function of extracellular DA concentration $D_{\text{ext}}(t)$ (in nM):

$$\tau_n \frac{dV}{dt} = -V + g(D_{\text{ext}}) \cdot I_{\text{base}} \quad (5)$$

where $V(t)$ is the neural response (arbitrary units), $\tau_n = 0.02$ hours (≈ 72 seconds) is the membrane time constant, $I_{\text{base}} = 1.0$ is a constant baseline drive, and the dopamine-dependent gain is modelled as a saturating Hill function:

$$g(D) = g_0 \left(1 + \alpha \frac{D}{K_D + D} \right) \quad (6)$$

with $g_0 = 1.0$ (baseline gain), $\alpha = 2.0$ (modulation strength), and $K_D = 2.0$ nM (half-maximal concentration). This form is motivated by the pharmacological observation that dopamine receptor activation follows binding kinetics with eventual saturation at high DA concentrations [15]. At normal baseline DA (~ 2.6 nM), the gain is approximately 2.1, and it can range from $g_0 = 1$ (zero DA) to $g_0(1 + \alpha) = 3$ at saturating concentrations.

The key quantities of interest are the *peak gain* reached during an LD dose (relevant to dyskinesia: excessive gain excursions are associated with supra-threshold dopaminergic stimulation), and the *gain variability* (max minus min during the dose window), which reflects how unstable the circuit becomes.

5.2 Results and Interpretation

Figure 6 shows the full coupled dynamics for four degeneration levels during an LD dose. The top panel is extracellular DA, the middle panel is the resulting gain $g(D_{\text{ext}})$, and the bottom panel is the neural response $V(t)$.

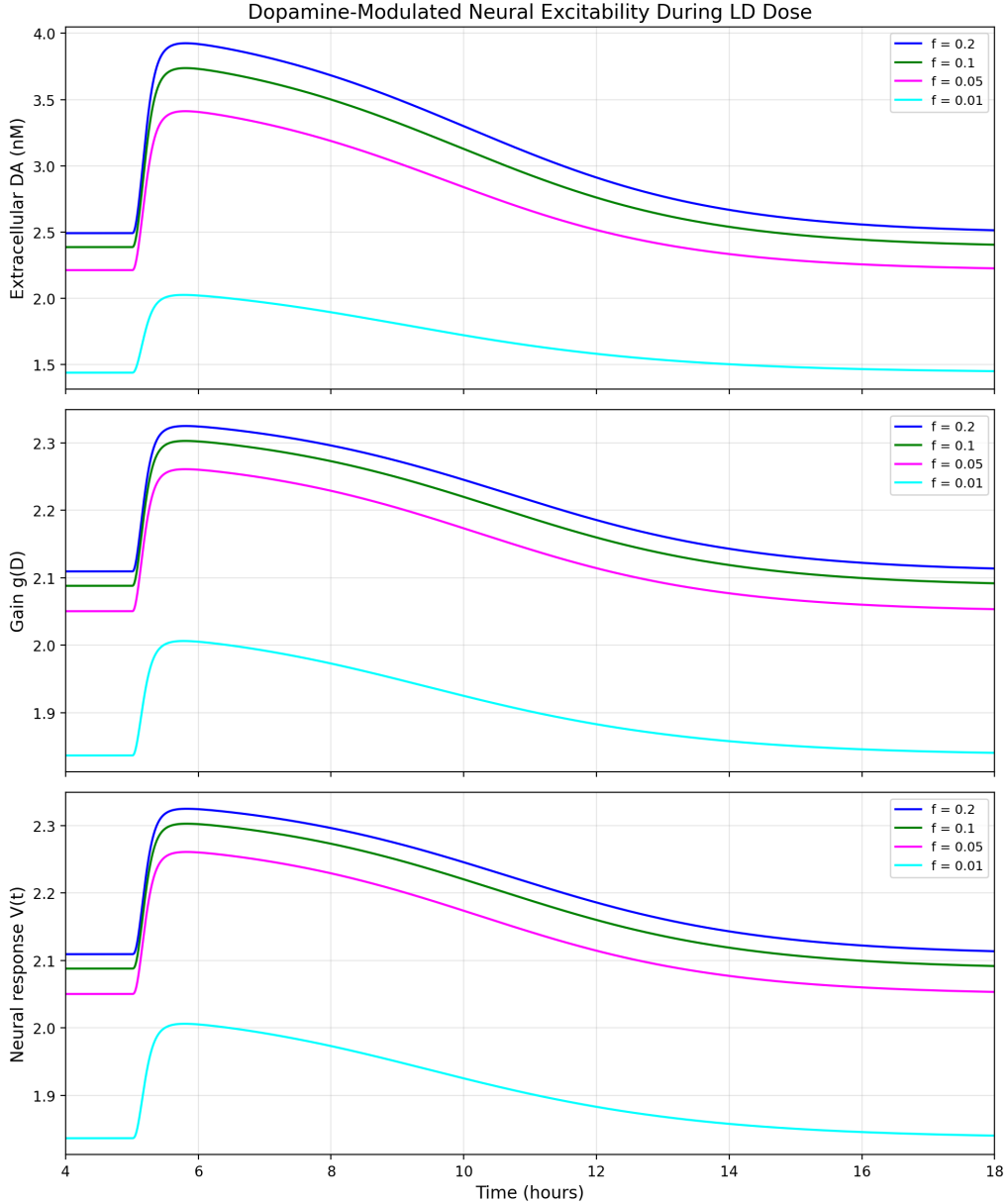


Figure 6: Dopamine-modulated neural excitability during LD dosing at four degeneration levels. **Top:** extracellular DA. **Middle:** dopamine-dependent gain. **Bottom:** downstream neural response. The $f = 0.01$ case (cyan) shows persistently low gain with minimal LD response, while $f = 0.2$ (blue) produces the largest transient gain excursion.

Several things are worth noting. For $f = 0.2$ and $f = 0.1$, the gain excursion during the LD dose is largest, reflecting the non-linear interplay between high DA peaks and the saturating gain function. In these intermediate degeneration cases, LD produces the sharpest and most variable gain dynamics, with a rapid rise followed by a rapid fall back to baseline. This is precisely the kind of all-or-nothing, oscillatory neural gain profile that is thought to underlie motor fluctuations and dyskinesias clinically [6]. For $f = 0.01$, the gain remains persistently low throughout the dose; there is barely any detectable response, reflecting the inability of a severely depleted system to deliver enough DA to drive meaningful changes in neural excitability.

Figure 7 quantifies these trends across the full range of f . Both peak gain and gain variability follow a non-monotonic pattern, peaking at intermediate degeneration ($f \approx 0.1$ – 0.2) and decreasing toward both the healthy and severely degenerated ends of the spectrum. For a healthy individual ($f \approx 1$), DAT-mediated buffering suppresses large DA excursions and the gain stays relatively stable. For severe degeneration ($f \ll 0.1$), the DA peaks themselves are small because the high removal rate clears DA faster than it can accumulate, so the gain excursions are also limited. The intermediate regime is in some sense the worst of both worlds: the terminal density is low enough to disable buffering, but high enough to still produce substantial DA peaks when LD is administered, resulting in large and rapid gain fluctuations in downstream circuits.

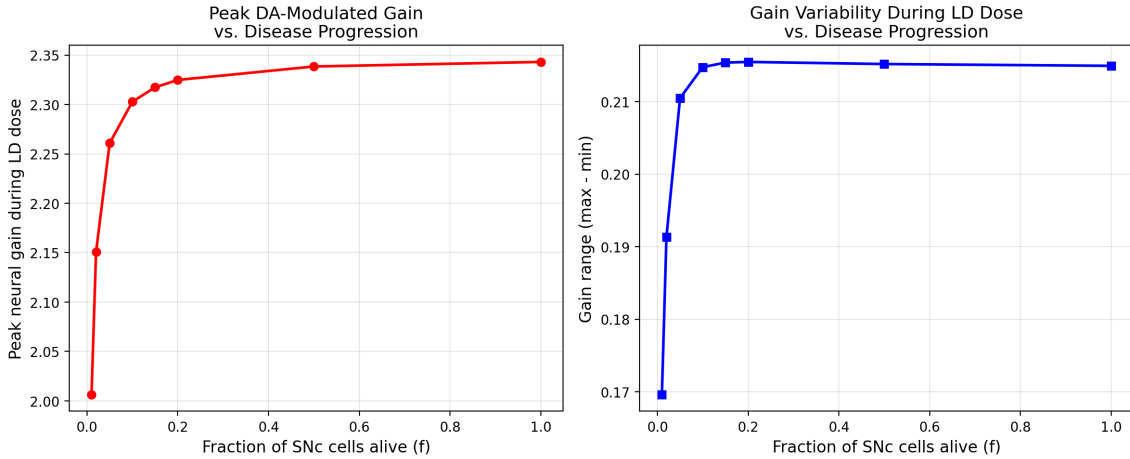


Figure 7: Left: Peak neural gain during LD dose as a function of f . **Right:** Gain variability (range of gain during dose period). Both show non-monotonic dependence on degeneration, with maximal circuit instability at intermediate disease stages.

This non-monotonic gain profile as a function of f is the most important result of the extension. It identifies a “danger zone” of degeneration (roughly $f = 0.05$ – 0.2) where LD therapy produces the largest and most variable gain excursions in downstream neurons, which is exactly the clinical stage at which dyskinesias tend to emerge [6]. Importantly, the model attributes these fluctuations to a structural property of the biochemical system—the mismatch between partial DA release from the remaining terminals plus serotonergic contribution and the declining clearance capacity—rather than to intrinsic changes in post-synaptic receptor expression, though both are likely involved in practice.

6 Discussion

6.1 Summary

This project reproduced the core published results of Best et al. [1] and Reed et al. [2] through an independent Python implementation: steady-state homeostasis, substrate inhibition of TH, passive stabilization, LD dose-dependent DA dynamics at varying degeneration levels, and therapeutic window shortening. The novel extension to dopamine-modulated neural excitability revealed a non-monotonic gain variability profile as a function of disease progression, offering a mechanistic rationale for why dyskinesias emerge preferentially at intermediate degeneration stages and why LD therapy loses effectiveness in advanced disease.

6.2 Comparison with Published Results

The reproduced results are qualitatively consistent with the published figures throughout. The passive stabilization curve in Figure 3 shows the expected plateau followed by decline, matching the shape of Reed et al. [2] Figure 6. The LD dose time courses in Figure 4 show the correct ordering of peak heights and the progressive narrowing of the therapeutic window. The therapeutic window curve in Figure 5 demonstrates the expected monotonic decline from roughly 11 hours to near zero, consistent with the published range. Quantitative differences, especially in absolute DA concentrations, reflect sensitivity of the bipterin cycling and autoreceptor formulation to exact parameter choices.

6.3 Limitations

This re-implementation focuses on the dopaminergic terminal subsystem and does not include the serotonergic terminal and raphe nucleus cell body models from Reed et al. [2]. In the full published model, serotonergic neurons become the dominant source of striatal DA release in advanced PD because they also take up LD via AADC and release the resulting DA without autoreceptor regulation, which is a key driver of peak DA heights in the severely degenerated states and a proposed mechanism for dyskinesias. Because the serotonergic subsystem is not included here, the DA peaks for low f values in Figure 4 do not exhibit the pronounced rise-then-fall shape seen in Reed et al. [2] Figure 7A, and peak heights at severe degeneration are likely underestimated as a result. Additionally, the neural excitability extension is intentionally simplified and should be read as illustrative rather than biophysically complete; a more rigorous treatment would couple the DA dynamics to a conductance-based model of medium spiny neurons with explicit D1 and D2 receptors that better capture the asymmetry between the direct and indirect pathways that underlies PD motor symptoms. The model also assumes a well-mixed extracellular space and does not capture spatial diffusion gradients or stochastic vesicle release, both of which may matter at low terminal densities.

6.4 Future Work

A few natural extensions stand out. The most immediate would be incorporating the full serotonergic subsystem from Reed et al. [2], which would restore the rise-then-fall DA peak pattern

at low f values that this implementation underestimates and would make the therapeutic window predictions more quantitatively accurate. On the neural side, replacing the leaky integrator with a conductance-based model of striatal medium spiny neurons with explicit D1 and D2 receptor pharmacology would allow more direct comparison with electrophysiological data and better capture the asymmetry between the direct and indirect pathways that underlies PD motor symptoms. Finally, the gain variability metric introduced here could potentially be used as an objective function to optimise LD dosing schedules—identifying the timing and dose magnitude that minimises gain fluctuations across the therapeutic window, which is essentially what continuous dopamine delivery strategies already aim for clinically [8].

7 Conclusion

Parkinson’s disease is not simply a dopamine deficit—it is a progressive loss of the system’s ability to regulate and buffer dopamine fluctuations, and that distinction matters for how we think about treatment. The homeostatic mechanisms that normally keep extracellular DA stable (passive stabilization, autoreceptor feedback, and DAT-mediated clearance) all degrade together as degeneration advances, converting a well-buffered system into one that is increasingly sensitive to pharmacological perturbation. The results here reinforce that picture: passive stabilization maintains near-normal DA until more than 80% of SNc neurons are gone, the therapeutic window shrinks from roughly 11 hours to near the plasma half-life of LD itself, and the novel extension shows that the most unstable regime is not end-stage disease but the intermediate range ($f \approx 0.1\text{--}0.2$) where the system still responds to levodopa but can no longer contain the resulting fluctuations—precisely the stage at which dyskinesias tend to emerge clinically [6]. Mathematical modelling is what makes it possible to see that non-monotonic structure clearly, and future work building on this framework has real potential to inform smarter, more personalised approaches to levodopa therapy.

Appendix A: Full ODE System: Best et al. (2009)

The nine coupled ODEs governing the dopaminergic terminal are listed below for completeness. The velocity functions (V_{TH} , V_{DRR} , V_{AADC} , V_{MAT} , V_{DAT} , V_{catab}^{eda}) and parameter values are described in Sections 2.2 and 3.1; this is just the full system in one place for reference.

$$\frac{d[bh_2]}{dt} = V_{TH} - V_{DRR} \quad (7)$$

$$\frac{d[bh_4]}{dt} = V_{DRR} - V_{TH} \quad (8)$$

$$\frac{d[tyr]}{dt} = V_{tyr,in} - V_{TH} - k_1[tyr] + k_{-1}[tyrpool] - k_{catab}^{tyr}[tyr] \quad (9)$$

$$\frac{d[l dopa]}{dt} = V_{TH} - V_{AADC} \quad (10)$$

$$\frac{d[cda]}{dt} = V_{AADC} - V_{MAT} + V_{DAT} - k_{catab}^{cda}[cda] \quad (11)$$

$$\frac{d[vda]}{dt} = V_{MAT} - k_{fire}[vda] \quad (12)$$

$$\frac{d[eda]}{dt} = k_{fire}[vda] - V_{DAT} - V_{catab}^{eda} - k_{rem}[eda] \quad (13)$$

$$\frac{d[hva]}{dt} = k_{catab}^{cda}[cda] + V_{catab}^{eda} - k_{catab}^{hva}[hva] \quad (14)$$

$$\frac{d[tyrpool]}{dt} = k_1[tyr] - k_{-1}[tyrpool] - k_{catab}^{tyrpool}[tyrpool] \quad (15)$$

All concentrations are in μM and time is in hours. k_{fire} is the vesicular release rate (tonic value 1/hr), and $k_{rem} = k_0/f$ in the Reed et al. [2] extension as described in Section 2.3.

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